

Hue, Saturation & Value

The three dimensions of color

Every color has three independent properties: hue (which color), saturation (how pure), and value (how light or dark). Understanding these three independently is the foundation of all color knowledge.

KEY CONCEPTS

Hue

Hue is the pure color identity — red, orange, yellow, green, blue, violet. The full spectrum of hues forms a color wheel. Hues that sit opposite each other on the wheel are complementary; adjacent hues are analogous.

Saturation

Saturation (chroma) measures how pure or grey a color is. Full saturation = the most vivid version of that hue. Zero saturation = pure grey. Most colors in nature are relatively desaturated. High saturation should be used sparingly for emphasis.

Value

Value is the lightness or darkness of a color, independent of its hue. Yellow is naturally high-value; violet is naturally low-value. You can have a dark, saturated red or a light, desaturated red — these are different values of the same hue.

Adjusting Independently

Practice adjusting one property while holding the others constant. This builds understanding of how each dimension works. When a color looks 'wrong', ask: is the hue wrong? The saturation? The value?

PRACTICE EXERCISES

1. Create a hue wheel with 12 colors from primary mixing alone.
2. Take one hue and make a 7-step saturation scale (full → grey).
3. Mix a 5-step value scale with one hue (dark → light without changing hue).

FURTHER READING

■ Ctrl+Paint – Color Theory Basics

<https://www.ctrlpaint.com/videos/color-theory>

■ Proko – Understanding Color

<https://youtu.be/AZhO4VjQUPE>